Thiemokho Fall | Applied AI & Scientific Machine Learning Researcher

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Profile

PhD candidate in **Applied Mathematics**, specializing in **Mathematical Modeling and Scientific Computing**, with an interdisciplinary background in **Artificial Intelligence, Machine Learning, Deep Learning and data-driven scientific modeling**.

My work combines **mathematical modeling, machine learning and scientific data analysis** to address complex real-world problems. My research experience includes **symbolic regression, explainable machine learning, deep learning, spatial modeling, multi-source data fusion and uncertainty-aware prediction**.

Strong interest in **Scientific Machine Learning, AI for complex systems, model discovery, optimization and data-driven modeling**, with practical experience developing Python-based research workflows and translating scientific methods into computational applications.

Core Expertise

Scientific AI: Scientific Machine Learning, Machine Learning, Deep Learning, Explainable AI, Symbolic Regression, Model Discovery, Uncertainty-Aware Prediction.

Mathematical Modeling: Mathematical modeling, numerical analysis, optimization, statistics, spatial modeling, geostatistics, data assimilation.

Machine Learning: Supervised learning, regression, anomaly detection, model evaluation, feature engineering, spatio-temporal modeling.

Programming: Python, SQL, R, Bash; scientific computing and reproducible research workflows.

Data Engineering: Data processing, ETL, scientific data pipelines, PostgreSQL, Docker, Apache Airflow, AWS S3/EC2.

Geospatial AI: Remote sensing, multi-source data fusion, GeoPandas, Rasterio, Google Earth Engine, QGIS.

Selected Research Experience

AIMS Senegal — Cooperative Master in Mathematical Sciences
Symbolic Regression & Explainable Machine Learning *2024–2026*

- Investigated **symbolic regression** for discovering analytical relationships between physical quantities directly from data, without imposing an a priori functional form.
- Applied **genetic programming** using the **gplearn** framework to evolve symbolic mathematical expressions through selection, crossover and mutation.
- Studied the trade-off between **prediction accuracy and expression complexity**, with constraints designed to obtain interpretable analytical models.
- Demonstrated that constrained genetic programming can recover **physically relevant mathematical relationships** from data.

MESRI — ESPOIR-Jeunes / Competitive Research Funding **Senegal**
MINERALS-AI Senegal — Scientific Coordinator & Operational Lead *2026–2027*

- Coordinate an interdisciplinary research project combining **Artificial Intelligence, mathematical modeling and multi-source scientific data**.
- Develop machine learning and deep learning approaches for complex heterogeneous datasets.
- Work on **data fusion, spatial modeling, uncertainty analysis and predictive modeling**.
- Coordinate research activities and contribute to the development of a prototype decision-support system.

PanAfrican Mineral Resources (PMR) **Dakar, Senegal**
AI-based Data Science for Complex Scientific Systems *2025*

- Developed Python-based workflows for heterogeneous scientific datasets.
- Applied **Machine Learning to anomaly detection and predictive modeling**.
- Integrated geological, geochemical, geophysical and satellite-derived data.
- Developed computational workflows for data processing, feature engineering and spatial analysis.

Professional Experience

Centre de Suivi Écologique (CSE) **Dakar, Senegal**
Data Scientist / Geospatial AI Research Intern *Oct. 2025–Apr. 2026*

- Developed Python and Google Earth Engine workflows for large-scale environmental and Earth observation datasets.
- Applied **Machine Learning, Deep Learning and spatio-temporal modeling** to environmental applications.
- Integrated heterogeneous observations from satellite and environmental data sources.
- Contributed to SERVIR-related Earth observation activities supported by NASA and USAID.
- Delivered training and knowledge transfer in Python, remote sensing and geospatial data analysis.

Selected Scientific Project

Centre de Suivi Écologique **Senegal / Sahel**
Spatio-temporal Modeling of Sahelian Vegetation Dynamics *2025–2026*

- Analysed multi-decadal environmental data covering **1984–2025**.
- Integrated Landsat NDVI, CHIRPS, ERA5-Land and TerraClimate datasets.
- Developed and evaluated **ConvLSTM models** for spatio-temporal vegetation dynamics.
- Applied clustering and spatial analysis to identify regional patterns.
- Investigated relationships between climate variables and vegetation dynamics.

Education

Cheikh Anta Diop University (UCAD) **Dakar, Senegal**
PhD in Applied Mathematics — Mathematical Modeling and Scientific Computing *Ongoing*
Research areas: **mathematical modeling, data assimilation, optimization under uncertainty, spatial modeling and Artificial Intelligence**.

AIMS Senegal
Cooperative Master in Mathematical Sciences — Data Science *2024–2026*
Training in **Machine Learning, Deep Learning, NLP, Generative AI and Big Data**. **Master's thesis:** Symbolic Regression and Explainable Machine Learning.

Cheikh Anta Diop University (UCAD) **Dakar, Senegal**
Master 2 in Algebra and Applications *2025*
Mathematical research involving algebraic methods, Gröbner bases and applications.

Technical Skills

Programming: **Python**, SQL, R, Bash

AI / ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Genetic Programming, Symbolic Regression

Scientific Computing: NumPy, Pandas, mathematical modeling, numerical methods, optimization, statistics

Data Engineering: PostgreSQL, Docker, Apache Airflow, AWS S3/EC2, ETL pipelines

Geospatial: GeoPandas, Rasterio, Google Earth Engine, geemap, QGIS, WhiteboxTools

Development: FastAPI, Streamlit, Git, GitHub, Linux, LaTeX

Scientific Contributions

5th International Conference on Modeling (CIM), UCAD
Multi-source Geoscientific Data Fusion, Deep Learning (FCN) and Spatially Constrained Clustering for Gold Potential Mapping in the Senegalese Basement (Mako Area).
Second author and presenter.

Teaching & Knowledge Transfer

Cheikh Anta Diop University (UCAD) **Dakar, Senegal**
Teaching Assistant / Adjunct Lecturer *2023–2026*
Teaching mathematics, algebra, linear algebra, Python and applied mathematics.

Workshops and training activities in **Python, remote sensing, GIS and geospatial data analysis**.

Languages

- **French:** Fluent
- **English:** Professional working proficiency

References

Prof. Mouhamadou Bamba Sylla — AIMS-Network Research Chair in Climate Change Science, AIMS Research and Innovation Centre, Kigali, Rwanda
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